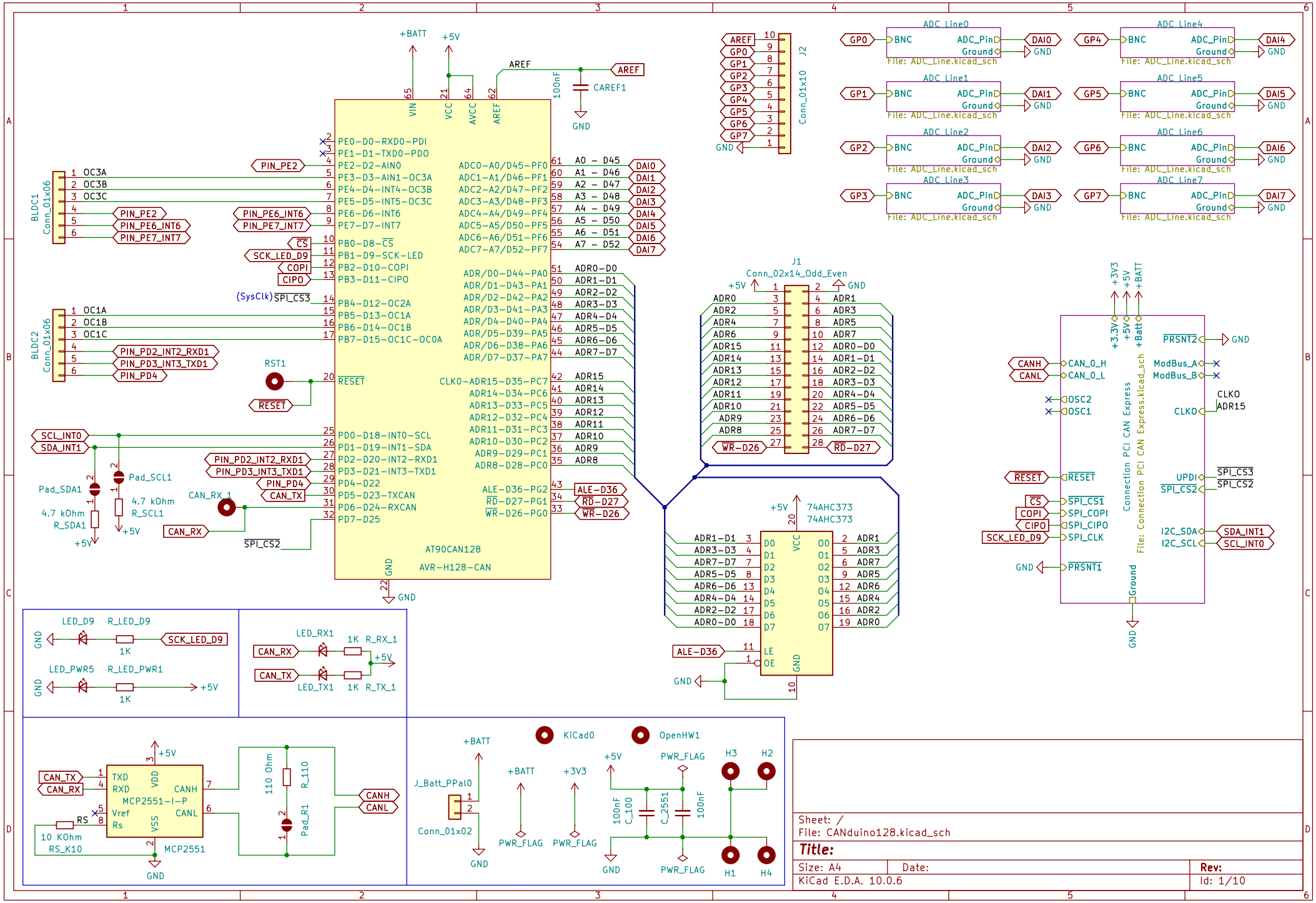
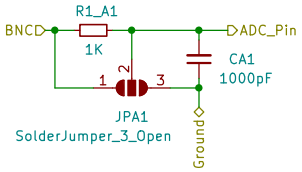
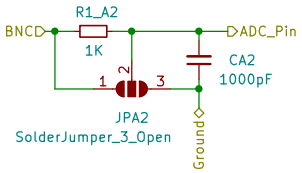
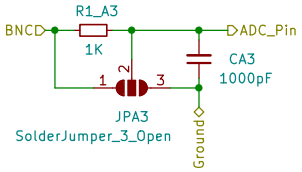

+BATT
+5V
AREF
CAREF1
100nF
GND
AT90CAN128
AVR-H128-CAN
PE0-D0-RXD0-PDI
PE1-D1-TXD0-PDO
PE2-D2-AIN0
PE3-D3-AIN1-OC3A
PE4-D4-INT4-OC3B
PE5-D5-INT5-OC3C
PE6-D6-INT6
PE7-D7-INT7
PB0-D8-CS
PB1-D9-SCK_LED
PB2-D10-COPI
PB3-D11-CIPO
PB4-D12-OC2A
PB5-D13-OC1A
PB6-D14-OC1B
PB7-D15-OC1C-OC0A
PD0-D18-INT0-SCL
PD1-D19-INT1-SDA
PD2-D20-INT2-RXD1
PD3-D21-INT3-TXD1
PD4-D22
PD5-D23-TXCAN
PD6-D24-RXCAN
PD7-D25
ADC0-A0/D45-PF0
ADC1-A1/D46-PF1
ADC2-A2/D47-PF2
ADC3-A3/D48-PF3
ADC4-A4/D49-PF4
ADC5-A5/D50-PF5
ADC6-A6/D51-PF6
ADC7-A7/D52-PF7
ADR0/D0-D44-PA0
ADR/D1-D43-PA1
ADR/D2-D42-PA2
ADR/D3-D41-PA3
ADR/D4-D40-PA4
ADR/D5-D39-PA5
ADR/D6-D38-PA6
ADR/D7-D37-PA7
CLKO-ADR15-D35-PC7
ADR14-D34-PC6
ADR13-D33-PC5
ADR12-D32-PC4
ADR11-D31-PC3
ADR10-D30-PC2
ADR9-D29-PC1
ADR8-D28-PC0
ALE-D36-PG2
RD-D27-PG1
WR-D26-PG0
RESET
BLDC1
BLDC2
OC3A
OC3B
OC3C
OC1A
OC1B
OC1C
PIN_PE2
PIN_PE6_INT6
PIN_PE7_INT7
PIN_PD2_INT2_RXD1
PIN_PD3_INT3_TXD1
PIN_PD4
CS
SCK_LED_D9
COPI
CIPO
SPI_CS3
SPI_CS2
RST1
SCL_INT0
SDA_INT1
Pad_SDA1
Pad_SCL1
4.7 kOhm
R_SDA1
R_SCL1
CAN_RX_1
CAN_RX
CAN_TX
ALE-D36
RD-D27
WR-D26
J2
Conn_01x10
J1
Conn_02x14_Odd_Even
74AHC373
ADR0
ADR1
ADR2
ADR3
ADR4
ADR5
ADR6
ADR7
ADR8
ADR9
ADR10
ADR11
ADR12
ADR13
ADR14
ADR15
ADR0-D0
ADR1-D1
ADR2-D2
ADR3-D3
ADR4-D4
ADR5-D5
ADR6-D6
ADR7-D7
ADC_Line0
ADC_Line1
ADC_Line2
ADC_Line3
ADC_Line4
ADC_Line5
ADC_Line6
ADC_Line7
BNC
ADC_Pin
Ground
File: ADC_Line.kicad_sch
GP0
GP1
GP2
GP3
GP4
GP5
GP6
GP7
DAI0
DAI1
DAI2
DAI3
DAI4
DAI5
DAI6
DAI7
+3V3
+Batt
PRSNT2
PRSNT1
CANH
CANL
CAN_0_H
CAN_0_L
ModBus_A
ModBus_B
OSC2
OSC1
CLKO
UPDI
I2C_SDA
I2C_SCL
Connection PCI CAN Express
File: Connection PCI CAN Express.kicad_sch
LED_D9
R_LED_D9
1K
LED_PWR5
R_LED_PWR1
LED_RX1
1K R_RX_1
LED_TX1
1K R_TX_1
MCP2551-I-P
TXD
RXD
Vref
Rs
VDD
VSS
CANH
CANL
RS
10 KOhm
RS_K10
MCP2551
110 Ohm
R_110
Pad_R1
J_Batt_PPal0
Conn_01x02
PWR_FLAG
KiCad0
OpenHW1
C_100
C_2551
H1
H2
H3
H4
Sheet: /
File: CANduino128.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 1/10



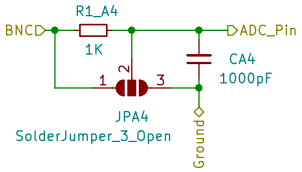
Sheet: /ADC_Line0/		
File: ADC_Line.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 2/10



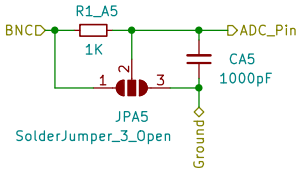
Sheet: /ADC_Line1/ File: ADC_Line.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 3/10



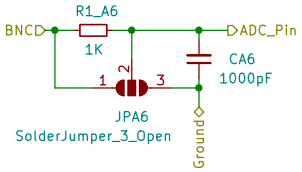
Sheet: /ADC_Line2/ File: ADC_Line.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 4/10



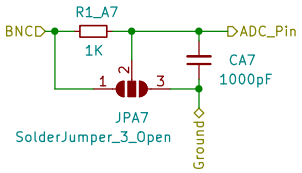
Sheet: /ADC_Line3/ File: ADC_Line.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 5/10



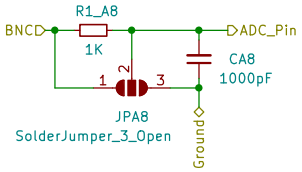
Sheet: /ADC_Line4/ File: ADC_Line.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 6/10



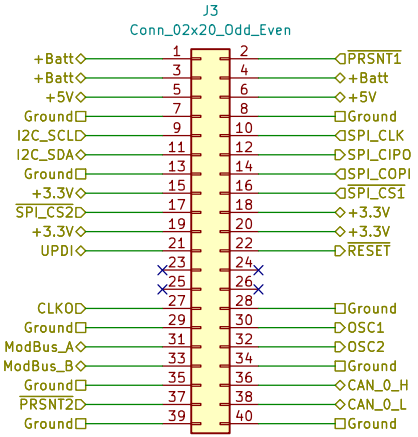
Sheet: /ADC_Line5/ File: ADC_Line.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 7/10

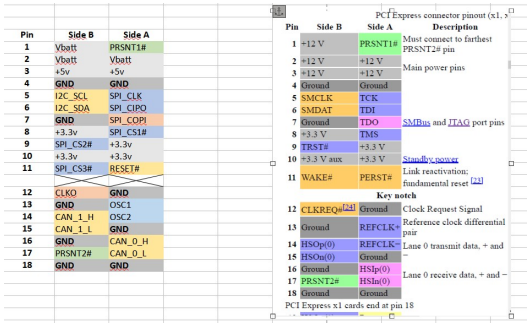


Sheet: /ADC_Line6/ File: ADC_Line.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 8/10



Sheet: /ADC_Line7/ File: ADC_Line.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 9/10





Pin	Side B	Side A
1	Vbatt	PRSNT1#
2	Vbatt	Vbatt
3	+5v	+5v
4	GND	GND
5	I2C_SCL	SPI_CLK
6	I2C_SDA	SPI_CIPO
7	GND	SPI_COPI
8	+3.3v	SPI_CS1#
9	SPI_CS2#	+3.3v
10	+3.3v	+3.3v
11	SPI_CS3#	RESET#
12	CLKO	GND
13	GND	OSC1
14	CAN_1_H	OSC2
15	CAN_1_L	GND
16	GND	CAN_0_H
17	PRSNT2#	CAN_0_L
18	GND	GND

PCI Express connector pinout (x1, x

Pin	Side B	Side A	Description
1	+12 V	PRSNT1#	Must connect to farthest PRSNT2# pin
2	+12 V	+12 V	Main power pins
3	+12 V	+12 V	
4	Ground	Ground	
5	SMCLK	TCK	
6	SMDAT	TDI	
7	Ground	TDO	SMBus and JTAG port pins
8	+3.3 V	TMS	
9	TRST#	+3.3 V	
10	+3.3 V aux	+3.3 V	Standby power
11	WAKE#	PERST#	Link reactivation; fundamental reset [23]
	Key notch		
12	CLKREQ# [24]	Ground	Clock Request Signal
13	Ground	REFCLK+	Reference clock differential pair
14	HSOp(0)	REFCLK−	Lane 0 transmit data, + and −
15	HSOn(0)	Ground	
16	Ground	HSIp(0)	
17	PRSNT2#	HSIn(0)	Lane 0 receive data, + and −
18	Ground	Ground	

PCI Express x1 cards end at pin 18

Liaison :
- PRSNT1# et PRSNT2# par le GNDD (équivalent présence carte PCIx1)

Remplacement :
- SMBus -> I2C
- JTAG -> SPI dont SC1# et SC2#
- WAKE# -> CS3#
- CLKREQ -> CLKO
- RefClk différentielle PCI -> OSC1 et OSC2 pour external clock MCU
- HSOp(0) soit TX0 PCI -> ModBus A+
- HSOn(0) soit TX0 PCI -> ModBus B-
- HSIp(0) soit RXO PCI -> CAN_0_H
- HSIn(0) soit RXO PCI -> CAN_0_L

PS :
L'utilisation d'un RS485 4 fils est possibles en cablant :
- HSOp(0) soit TX0 PCI -> DI_A+ master / RO_A+ slave(s)
- HSOn(0) soit TX0 PCI -> DI_B- master / RO_B- slave(s)
- HSIp(0) soit RXO PCI -> RO_A+ master / DI_A+ slave(s)
- HSIn(0) soit RXO PCI -> RO_B- master / DI_B- slave(s)

L'utilisation d'un RS485 2 fils est possibles en cablant :
- HSOp(0) soit TX0 PCI -> A+ master / A+ slave
- HSOn(0) soit TX0 PCI -> B- master / B- slave
- HSIp(0) soit RXO PCI -> NC. ou CAN_0_H
- HSIn(0) soit RXO PCI -> NC. ou CAN_0_L

Sheet: /Connection PCI CAN Express/
File: Connection PCI CAN Express.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 10/10